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CAT 1

Q1.

Check if the dot product of every distinct pair of vectors is zero ($V_i \cdot V_j = 0$ for $i \neq j$).

i. $V_1 \cdot V_2$:

$$\begin{aligned} & (2/3)(2/3) + (-2/3)(1/3) + (1/3)(-2/3) \\ &= 4/9 - 2/9 - 2/9 \\ &= 0 \end{aligned}$$

ii. $V_1 \cdot V_3$:

$$\begin{aligned} & (2/3)(1/3) + (-2/3)(2/3) + (1/3)(2/3) \\ &= 2/9 - 4/9 + 2/9 \\ &= 0 \end{aligned}$$

iii. $V_2 \cdot V_3$:

$$\begin{aligned} & (2/3)(1/3) + (1/3)(2/3) + (-2/3)(2/3) \\ &= 2/9 + 2/9 - 4/9 \\ &= 0 \end{aligned}$$

Since all the dot products are zero, the set is orthogonal

Now that we know the set is orthogonal, we check if each vector has a magnitude(norm) of 1

$$(\|v\| = \sqrt{v \cdot v} = 1)$$

$$\begin{aligned} \|v_1\|: & \sqrt{[(2/3)^2 + (-2/3)^2 + (1/3)^2]} \\ &= \sqrt{4/9 + 4/9 + 1/9} \\ &= \sqrt{9/9} \\ &= \sqrt{1} \\ &= 1 \end{aligned}$$

$$\begin{aligned} \|v_2\|: & \sqrt{[(2/3)^2 + (1/3)^2 + (-2/3)^2]} \\ &= \sqrt{4/9 + 1/9 + 4/9} \\ &= \sqrt{9/9} \\ &= \sqrt{1} \\ &= 1 \end{aligned}$$

$$\begin{aligned} \|v_3\|: & \sqrt{[(1/3)^2 + (2/3)^2 + (2/3)^2]} \\ &= \sqrt{1/9 + 4/9 + 4/9} \\ &= \sqrt{9/9} \\ &= \sqrt{1} \\ &= 1 \end{aligned}$$

Since all the $\|v_i\| = 1$, the set is orthonormal.

Q2.

Getting the augmented matrix $[A|I]$

$$\begin{bmatrix} 1 & 2 & 1 & | & 1 & 0 & 0 \\ 1 & 1 & 1 & | & 0 & 1 & 0 \\ 0 & 3 & 1 & | & 0 & 0 & 1 \end{bmatrix}$$

Performing row operations to get identity on the left

$$R_2 \leftarrow R_2 - R_1:$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & -1 & 1 & 0 \\ 0 & 3 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$R_2 \leftarrow -1 \times R_2 \text{ (Make pivot positive):}$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 3 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$R_3 \leftarrow R_3 - 3R_2:$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -3 & 3 & 1 \end{array} \right]$$

$$R_1 \leftarrow R_1 - R_3 \text{ (Eliminate top right):}$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 4 & -3 & -1 \\ 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -3 & 3 & 1 \end{array} \right]$$

$$R_1 \leftarrow R_1 - 2R_2:$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & -1 & -1 \\ 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -3 & 3 & 1 \end{array} \right]$$

Inverse of A

$$A^{-1} = \begin{bmatrix} 2 & -1 & -1 \\ 1 & -1 & 0 \\ -3 & 3 & 1 \end{bmatrix}$$

Q3.

Find Eigenvalues(λ): Solving $\det(A - \lambda I) = 0$

$$\det \begin{pmatrix} 3 - \lambda & 1 \\ 1 & 3 - \lambda \end{pmatrix} = (3 - \lambda)^2 - 1 = 0$$

$$(3 - \lambda)^2 = 1$$

$$3 - \lambda = \pm 1$$

$$\lambda_1 = 4, \lambda_2 = 2$$

Finding normalized Eigenvectors

For $\lambda_1 = 4$,

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x = y$$

Eigenvector $v_1 = (1, 1)$

$$\text{normalized } u_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$$

For $\lambda_2 = 2$,

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x = -y$$

Eigenvector $v_2 = (-1, 1)$

$$\text{normalized } u_2 = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$$

$$A = \lambda_1 u_1 u_1^T + \lambda_2 u_2 u_2^T$$

$$A = 4 \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} + 2 \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$= 4 \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} + 2 \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix}$$

Q4.

Let the columns of A be a_1, a_2, a_3 .

$$a_1 = (1, -1, 0)^T,$$

$$a_2 = (2, 2, 1)^T,$$

$$a_3 = (1, 2, 0)^T$$

$$u_1 = a_1 = (1, -1, 0)^T$$

$$e_1 = \frac{u_1}{\|u_1\|} = \frac{1}{\sqrt{2}} (1, -1, 0)^T$$

Orthogonalize a_2

$$u_2 = a_2 - \frac{a_2 \cdot u_1}{u_1 \cdot u_1} u_1$$

$$a_2 \cdot u_1 = 2 \cdot 1 + 2 \cdot (-1) + 1 \cdot 0 = 0$$

- Since the dot product = 0, a_2 is orthogonal to u_1 .

$$U_2 = (2, 2, 1)^T$$

$$\begin{aligned} e_2 &= \frac{u_2}{\|u_2\|} \\ &= \frac{1}{\sqrt{4+4+1}} (2, 2, 1)^T \\ &= \frac{1}{3} (2, 2, 1)^T \end{aligned}$$

Orthogonalize a_3

$$u_3 = a_3 - \text{proj}_{u_1} a_3 - \text{proj}_{u_2} a_3$$

$$\begin{aligned} \text{proj}_{u_1} a_3 &= \frac{1 \cdot 1 + 2 \cdot (-1) + 0}{2} u_1 = -\frac{1}{2} (1, -1, 0) \\ &= (-0.5, 0.5, 0) \end{aligned}$$

$$\begin{aligned} \text{proj}_{u_2} a_3 &= \frac{1 \cdot 2 + 2 \cdot 2 + 0 \cdot 1}{9} u_2 = \frac{2}{3} (2, 2, 1) \\ &= (1.33, 1.33, 0.67) \end{aligned}$$

$$u_3 = (1, 2, 0) - (-0.5, 0.5, 0) - (1.33, 1.33, 0.67)$$

$$u_3 = \left(\frac{1}{6}, \frac{1}{6}, -\frac{4}{6} \right)$$

Simplify direction: (1, 1, -4)

Normalize e_3

$$e_3 = \frac{1}{\sqrt{18}} (1, 1, -4) = \frac{1}{3\sqrt{2}} (1, 1, -4)$$

Constructing Q and R

$$Q=[e_1, e_2, e_3]=\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{2}{3} & \frac{1}{3\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{2}{3} & \frac{1}{3\sqrt{2}} \\ 0 & \frac{1}{3} & -\frac{4}{3\sqrt{2}} \end{bmatrix}$$

$$R=Q^T A=\begin{bmatrix} \|u_1\| & a_2 \cdot e_1 & a_3 \cdot e_1 \\ 0 & \|u_2\| & a_3 \cdot e_2 \\ 0 & 0 & \|u_3\| \end{bmatrix}$$

$$=\begin{bmatrix} \sqrt{2} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 3 & 2 \\ 0 & 0 & \frac{\sqrt{18}}{6} \end{bmatrix}$$

Q5.

Calculating the means

$$\bar{x}_1=\frac{4+4+8+12}{4}=7$$

$$\bar{x}_2=\frac{2+5+10+9}{4}=6.5$$

$$\bar{x}_3=\frac{0+6+12+6}{4}=6$$

centering the data $X_c=X-\bar{X}$

$$X_c=\begin{bmatrix} 4-7 & 2-6.5 & 0-6 \\ 4-7 & 5-6.5 & 6-6 \\ 8-7 & 10-6.5 & 12-6 \\ 12-7 & 9-6.5 & 6-6 \end{bmatrix}$$

$$=\begin{bmatrix} -3 & -4.5 & -6 \\ -3 & -1.5 & 0 \\ 1 & 3.5 & 6 \\ 5 & 2.5 & 0 \end{bmatrix}$$

Calculating the Covariance matrix

$$S=\frac{1}{n-1}X_c^T X_c$$

$$n = 4$$

$$\begin{aligned}s_{11} &= ((-3)^2 + (-3)^2 + 1^2 + 5^2)/3 \\ &= (9+9+1+25)/3 = 44/3 \\ &= 14.67\end{aligned}$$

$$\begin{aligned}s_{22} &= ((-4.5)^2 + (-1.5)^2 + 3.5^2 + 2.5^2)/3 \\ &= (20.25 + 2.25 + 12.25 + 6.25)/3 = 41/3 \\ &= 13.67\end{aligned}$$

$$\begin{aligned}s_{33} &= ((-6)^2 + 0 + 6^2 + 0)/3 \\ &= (36+36)/3 = 72/3 \\ &= 24\end{aligned}$$

$$\begin{aligned}s_{12} &= ((-3)(-4.5) + (-3)(-1.5) + (1)(3.5) + (5)(2.5))/3 \\ &= (13.5 + 4.5 + 3.5 + 12.5)/3 = 34/3 \\ &= 11.33\end{aligned}$$

$$\begin{aligned}s_{13} &= ((-3)(-6) + 0 + (1)(6) + 0)/3 \\ &= (18+6)/3 = 24/3 \\ &= 8\end{aligned}$$

$$\begin{aligned}s_{23} &= ((-4.5)(-6) + 0 + (3.5)(6) + 0)/3 \\ &= (27 + 21)/3 = 48/3 \\ &= 16\end{aligned}$$

$$S = \begin{bmatrix} 14.67 & 11.33 & 8 \\ 11.33 & 13.67 & 16 \\ 8 & 16 & 24 \end{bmatrix}$$

Q6.

Bayes Theorem

$$P(F|E) = \frac{P(E|F) \cdot P(F)}{P(E)}$$

$$P(F|E) = \frac{(2/5) \cdot (1/2)}{1/3}$$

$$\begin{aligned}&= \frac{2/10}{1/3} \\ &= 1/5 \times 3/1 \\ &= 3/5\end{aligned}$$

Q7.

Probabilities of picking blue

$$P(\text{Blue}|B_1) = 3/5$$

$$P(\text{Blue}|B_2) = 1/5$$

Total Probability of picking Blue $P(\text{Blue})$

$$P(\text{Blue}) = P(\text{Blue}|B_1)P(B_1) + P(\text{Blue}|B_2)P(B_2)$$

$$\begin{aligned}
 &= (3/5 \times 1/2) + (1/5 \times 1/2) \\
 &= 3/10 + 1/10 \\
 &= 2/5
 \end{aligned}$$

$$P(B_1|Blue)$$

$$= \frac{P(Blue|B_1)P(B_1)}{P(Blue)}$$

$$= \frac{3/10}{4/10}$$

$$= 3/4$$

$$= 0.75$$